INCH-POUND
MIL-I-24718/1
6 April 1990

MILITARY SPECIFICATION SHEET

INSULATING RESINS, SOLVENTLESS, VACUUM-PRESSURE-IMPREGNATING EPOXY-LEWIS ACID, SLIGHTLY THIXOTROPIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24718.

Resin: The product shall consist of primarily epoxy resin with a Lewis acid type catalyst and appropriate filler and thixotropes to satisfy the requirements specified.

Thermal class: The cured resin shall be of a thermal class as required by the contracting activity (see 4.6.15.1).

Rheology: The rheological properties shall be as specified and suitable for VPI of form and random wound electrical coils and windings.

Restrictions: The epoxy-Lewis acid insulating resin formulations are not suitable for use on high voltage ac equipment (>5KV).

First article inspection. First article inspection shall be as specified in MIL-I 24718 and table I herein.

TABLE I. First article inspection.

| Characteristics | Unit | Requirements |
|---|---|---|
| Catalyzed storage life | Months at 20°C, minimum | 9 |
| Vapor pressure | mm Hg, maximum | 0.5 |
| Weight loss on cure | Average percent, maximum | 3.0 |
| Flash point | °C, minimum | 93 |
| Viscosity | | |
| at 2 r/min | Average centipoise, maximum | 18,000 |
| at 20 r/min | Average centipoise, maximum | 8,000 |
| Shear thinning index | Minimum | 1.10 |
| Build | Mils, minimum | 1.5 |
| | Mils, maximum | 3.0 |

AMSC N/A FSC 5970

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TABLE I. First article inspection - Continued.

| Characteristics | Unit | Requirements |
|---|---|---|
| Resin retention | Weight percent, minimum | 75.0 |
| Hardness | Type D, minimum | 72 |
| Dielectric strength at 96/23/50 | Volts per mil, minimum average per panel | 1,000 |
| at 96/23/96 | Percent decrease from 96/23/50 value, maximum | 25 |
| at 24/23/water | Percent decrease from 96/23/50 value, maximum | 25 |
| Dissipation factor at 23°C | Percent, maximum | 3.0 |
| Dielectric constant at 23°C | Maximum | 5.0 |
| Bond strength at 23°C | Pounds force, minimum | 30.0 |
| at 150°C | Pounds force, minimum | 5.0 |
| Thermal endurance class 155 | °C at 20,000 hours | 155 |
| Saltwater proof | Hours, minimum | 100 |
| Hydrolytic stability | Percent decrease in average bond strength, maximum | 50 |
| Chemical resistance Hydraulic fluid | Percent decrease in average bond strength, maximum | 25 |
| Lubricating oil | Percent decrease in average bond strength, maximum | 25 |
| Cleaning fluid | Percent decrease in average bond strength, maximum | 25 |
| Distilled water | Percent decrease in average bond strength, maximum | 25 |
| Detergent solution | Percent decrease in average bond strength, maximum | 25 |
| Gel time | Minutes | Within manufacturer's specified range |

Quality conformance inspection. Quality conformance inspection shall be as specified in MIL-I-24718 and table II herein:

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TABLE II. Quality conformance requirements.

| Characteristics | Unit | Requirements |
|---|---|---|
| Weight loss on cure | Average percent, maximum | 3.0 |
| Gel time | Percentage of qualification value, plus or minus | 10 |
| Viscosity at 2 r/min | Average centipoise, maximum | 20,000 |
| Build | Percentage of qualification value, plus or minus | 15 |
| Resin retention | Weight percent, minimum | 75.0 |
| Specific gravity | Percentage of qualification value, plus or minus | 10.0 |
| Dielectric strength at 24/23/50 | Volts per mil, maximum | 1,000 |
| Hardness | Percentage of qualification value, plus or minus | 10.0 |
| Bond strength at 25°C | Pounds force, minimum | 30.0 |

Intended use: The product shall be used for VPI of form or random wound electrical coils. It shall be used only as new resin for starting-up a VPI process or replenishing stored VPI resin of the same formulation and from the same manufacturer.

Part or Identifying Number (PIN): The PIN shall be M24718/1-(dash number indicating thermal class). For example, M24718/1-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance.

Packaging: See MIL-I-24718 for packaging requirements.

Custodians:
Army - ER
Navy - SH
Air Force - 20

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
Army - ME
Navy - MC, AS

Preparing activity:
Navy - SH
(Project 5970-1002-01)



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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER: MIL-I-24718/1 | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|

3. DOCUMENT TITLE

INSULATING RESINS, SOLVENTLESS, VACUUM-PRESSURE-IMPREGNATING: GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
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| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME: Technical Point of Contact (TPOC): Mr. C.Y. Lu (SEA 56Z24)<br>PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS: | b. TELEPHONE (Include Area Code)<br>(1) Commercial: TPOC: 703-602-3123<br>(2) AUTOVON: 8-332-3123 |
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| c. ADDRESS (Include Zip Code)<br>Commander, Naval Sea Systems Command<br>Department of the Navy<br>Washington, DC 20362-5101 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

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